AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

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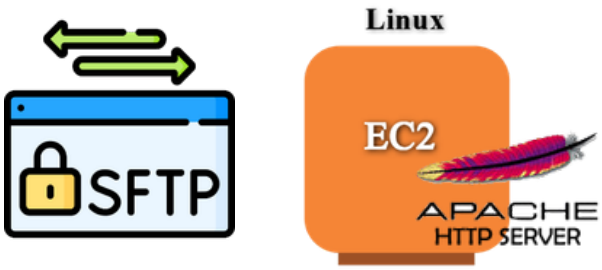
AWS Academy Certified Educator

7.Configure SFTP toLinux Web Server in AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.

  Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.

- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.

- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

SFTP- Secure File Transfer Protocol

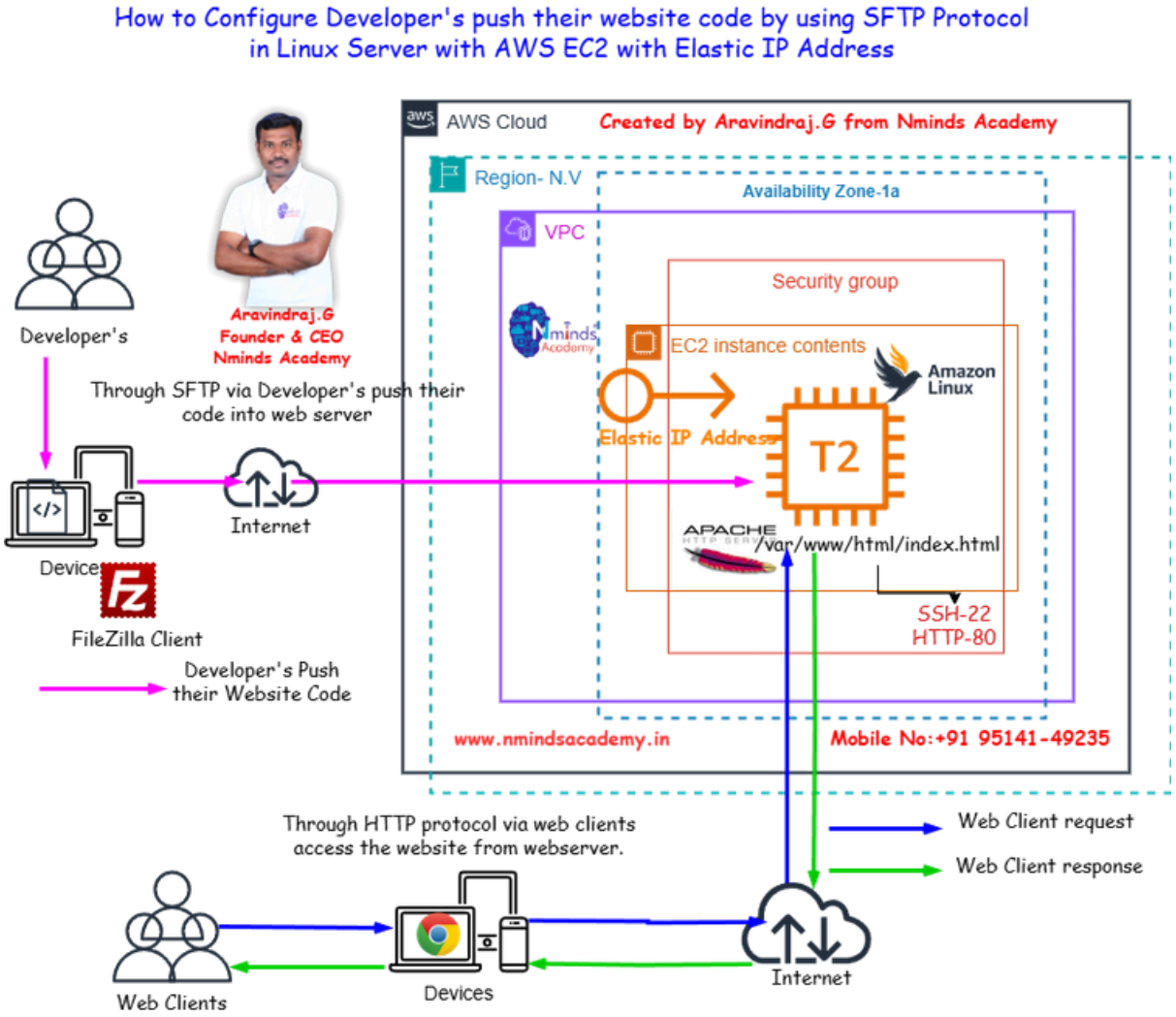
SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

Here are the key uses and advantages of SFTP:

1. File Transfer: SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. File Management: Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. Secure Authentication: SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. Encryption: The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.

5. Firewall-Friendly: SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.

Topology



Used Services in AWS

Amazon EC2

Used Services in Windows Server

APACHE HTTP SERVER

Apache Webserver

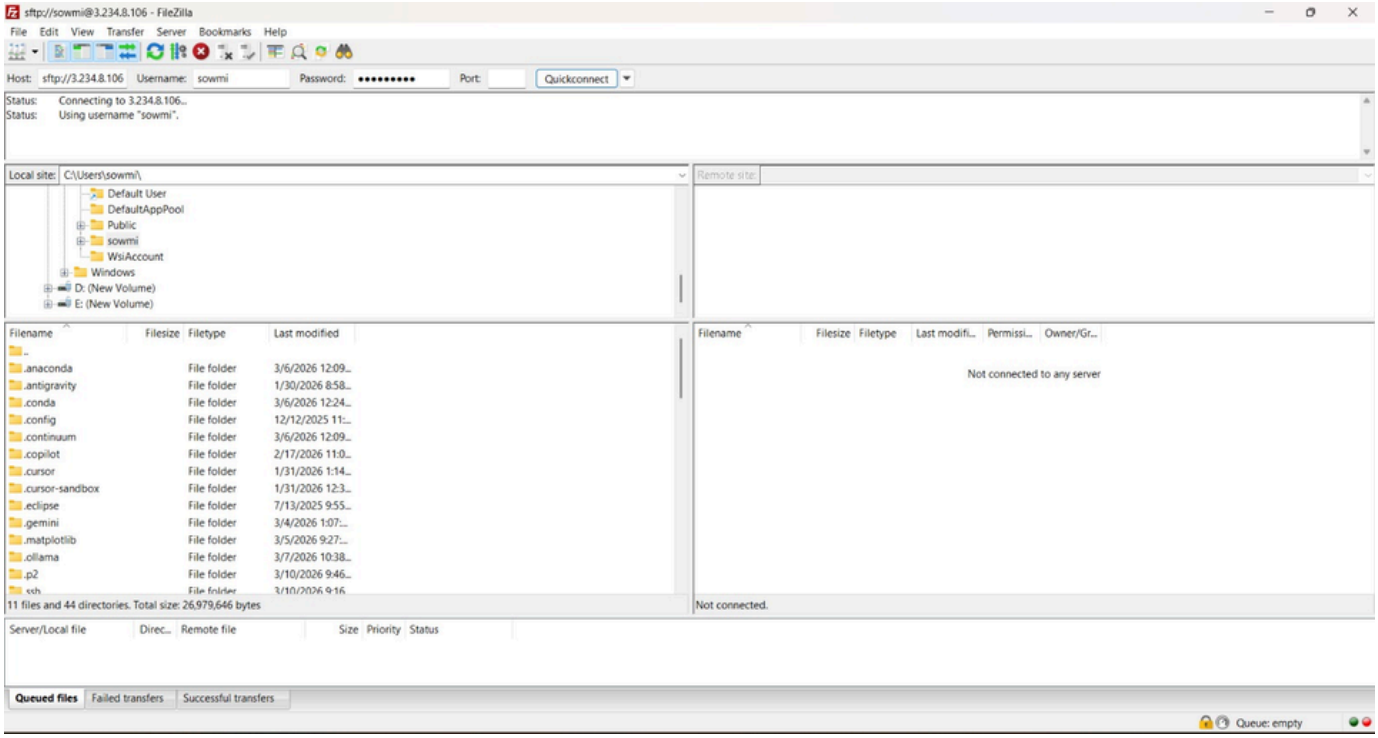
Amazon Elastic IP Address

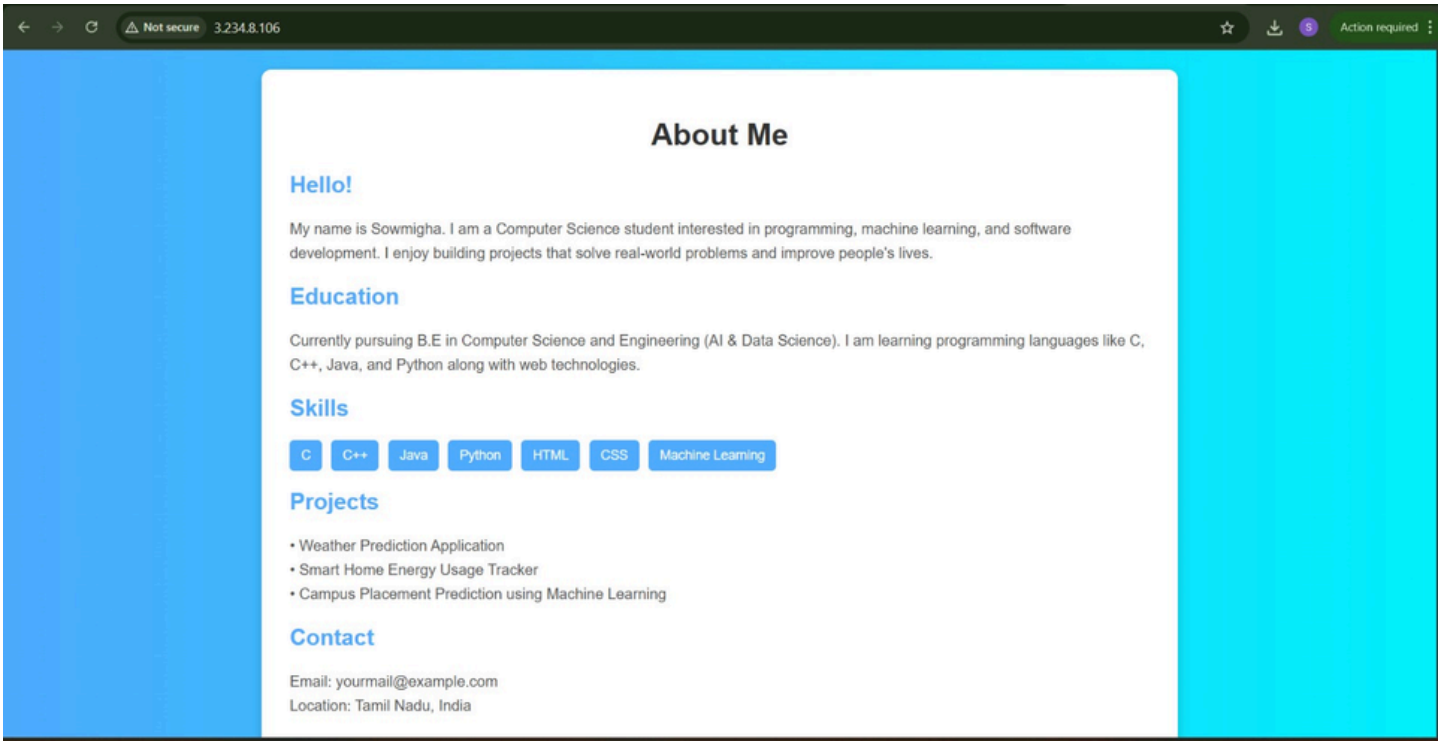
Execution Tasks:

```
root@ip-172-31-45-237:~
 Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
 Installing       : mod_lua-2.4.66-1.amzn2023.0.1.x86_64
 Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
 Installing       : httpd-2.4.66-1.amzn2023.0.1.x86_64
 Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64
 Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64
 Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64
 Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
 Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
 Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
 Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64
 Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64
 Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
 Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
 Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
 Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch
 Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
 Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64

nstalled:
 apr-1.7.5-1.amzn2023.0.4.x86_64                          apr-util-1.6.3-1.amzn2023.0.2.x86_64
 apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
 generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch        httpd-2.4.66-1.amzn2023.0.1.x86_64
 httpd-core-2.4.66-1.amzn2023.0.1.x86_64                  httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
 httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                 libbrotli-1.0.9-4.amzn2023.0.2.x86_64
 mailcap-2.1.49-3.amzn2023.0.3.noarch                     mod_http2-2.0.27-1.amzn2023.0.3.x86_64
 mod_lua-2.4.66-1.amzn2023.0.1.x86_64

omplete!
root@ip-172-31-45-237 ~]#
```

```
root@ip-172-31-45-237:~
 apr                          x86_64     1.7.5-1.amzn2023.0.4      amazonlinux     129 k
 apr-util                     x86_64     1.6.3-1.amzn2023.0.2      amazonlinux      97 k
 apr-util-lmdb                x86_64     1.6.3-1.amzn2023.0.2      amazonlinux      13 k
 generic-logos-httpd          noarch     18.0.0-12.amzn2023.0.3    amazonlinux      19 k
 httpd-core                   x86_64     2.4.66-1.amzn2023.0.1     amazonlinux     1.4 M
 httpd-filesystem             noarch     2.4.66-1.amzn2023.0.1     amazonlinux      13 k
 httpd-tools                  x86_64     2.4.66-1.amzn2023.0.1     amazonlinux      81 k
 libbrotli                    x86_64     1.0.9-4.amzn2023.0.2      amazonlinux     315 k
 mailcap                      noarch     2.1.49-3.amzn2023.0.3     amazonlinux      33 k
Installing weak dependencies:
 apr-util-openssl             x86_64     1.6.3-1.amzn2023.0.2      amazonlinux      15 k
 mod_http2                    x86_64     2.0.27-1.amzn2023.0.3     amazonlinux     166 k
 mod_lua                      x86_64     2.4.66-1.amzn2023.0.1     amazonlinux      60 k

Transaction Summary
================================================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                    3.5 MB/s | 129 kB     00:00
(2/13): apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64.rpm          313 kB/s |  13 kB     00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm               1.9 MB/s |  97 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm       654 kB/s |  15 kB     00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm  694 kB/s |  19 kB     00:00
(6/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm                 1.6 MB/s |  47 kB     00:00
(7/13): httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch.rpm      571 kB/s |  13 kB     00:00
(8/13): httpd-core-2.4.66-1.amzn2023.0.1.x86_64.rpm             30 MB/s | 1.4 MB     00:00
(9/13): httpd-tools-2.4.66-1.amzn2023.0.1.x86_64.rpm           2.4 MB/s |  81 kB     00:00
```

```
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:
ec2-user@ip-172-31-45-237:~
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\thanv>cd downloads

C:\Users\thanv\Downloads>ssh -i nmindsproject001.pem ec2-user@16.171.154.239
The authenticity of host '16.171.154.239 (16.171.154.239)' can't be established.
ED25519 key fingerprint is SHA256:mwYvXnoCaSNuYxD7EMRZI98Ry4IDYbEfGO1DhKXARXE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '16.171.154.239' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-45-237 ~]$
```


sftp://sowmi@3.234.8.106 - FileZilla
File Edit View Transfer Server Bookmarks Help
Host: sftp://3.234.8.106 Username: sowmi Password: ••••••••• Port: Quickconnect
Status: Connecting to 3.234.8.106...
Status: Using username "sowmi".
Local site: C:\Users\sowmi\
Remote site:
Default User
DefaultAppPool
Public
sowmi
WsiAccount
Windows
D: (New Volume)
E: (New Volume)
Filename Filesize Filetype Last modified
.anaconda File folder 3/6/2026 12:09...
.antigravity File folder 1/30/2026 8:58...
.conda File folder 3/6/2026 12:24...
.config File folder 12/12/2025 11:...
.continuum File folder 3/6/2026 12:09...
.copilot File folder 2/17/2026 11:0...
.cursor File folder 1/31/2026 1:14...
.cursor-sandbox File folder 1/31/2026 12:3...
.eclipse File folder 7/13/2025 9:55...
.gemini File folder 3/4/2026 1:07:...
.matplotlib File folder 3/5/2026 9:27:...
.ollama File folder 3/7/2026 10:38...
.p2 File folder 3/10/2026 9:46...
11 files and 44 directories. Total size: 26,979,646 bytes
Filename Filesize Filetype Last modifi... Permissi... Owner/Gr...
Not connected to any server
Not connected.
Server/Local file Direc... Remote file Size Priority Status
Queued files Failed transfers Successful transfers
Queue: empty


Not secure 3.234.8.106
Action required
About Me
Hello!
My name is Sowmigha. I am a Computer Science student interested in programming, machine learning, and software development. I enjoy building projects that solve real-world problems and improve people's lives.
Education
Currently pursuing B.E in Computer Science and Engineering (AI & Data Science). I am learning programming languages like C, C++, Java, and Python along with web technologies.
Skills
C
C++
Java
Python
HTML
CSS
Machine Learning
Projects
• Weather Prediction Application
• Smart Home Energy Usage Tracker
• Campus Placement Prediction using Machine Learning
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